
Adaptive Global–Local Fusion for Multi-category Protein–Protein Interaction Type Prediction using Graph Neural Networks

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1 Application domain and dataset

Protein–protein interactions (PPI) describe how proteins interact to perform biological functions. Accurate PPI prediction is important for understanding cellular processes, disease mechanisms, drug discovery, and disease treatment. Since traditional experimental methods for identifying PPI are often time-consuming and expensive, computational PPI prediction is an important large-scale biological application.

This project focuses on multi-category PPI interaction-type prediction. Unlike binary PPI prediction, which only determines whether two proteins interact, this task predicts the specific interaction categories between a protein pair. The problem naturally fits graph machine learning: proteins are nodes, protein–protein interactions are edges, and the target is the interaction type of each protein pair.

We use SHS27k and SHS148k, the benchmark datasets used in GNNGL-PPI. They are suitable because they provide multi-category PPI labels, support direct comparison with the original paper, and include three splitting strategies: Random, BFS, and DFS. These splits allow evaluation under both random and graph-structure-based generalization settings.

Table 1: Planned benchmark datasets.

Dataset	Number of proteins	Number of PPI pairs
SHS27k	1,690	7,624
SHS148k	5,189	44,488

Each protein pair may belong to one or more of seven categories:

Reaction, Binding, Ptmof, Activation, Inhibition, Catalysis, Expression.

Therefore, the label of each protein pair is a seven-dimensional multi-label vector.

2 Task formulation and evaluation

We model the PPI network as a graph:

$$G = (V, E),$$

where V is the set of proteins and E is the set of protein–protein interactions. Each protein $v \in V$ has an input feature vector x_v . For each protein pair (u, v) , the target label is:

$$y_{uv} \in \{0, 1\}^7.$$

The task is formulated as edge-level multi-label classification:

$$f(u, v, G, X) \rightarrow \hat{y}_{uv} \in [0, 1]^7.$$

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This formulation follows the course concepts of graph representation learning and edge-level prediction: a GNN encoder learns node representations from graph structure and node attributes, and a prediction head combines two node embeddings to predict the label of their interaction.

The primary metric is F1-measure, following the original GNNGL-PPI evaluation. Since the interaction categories are imbalanced, we will also report Micro-F1, Macro-F1, and Per-class F1. Macro-F1 and per-class F1 are useful for analyzing rare categories such as Expression, which may be underrepresented in overall metrics.

3 Baseline model: GNNGL-PPI

We use GNNGL-PPI as the baseline graph ML model. It directly targets multi-category PPI prediction and learns protein representations from both protein features and PPI network structure. Following the original paper, the model uses MASSA protein representations as node features. In the released implementation, these are loaded as pre-computed 512-dimensional embeddings rather than generated inside the repository.

GNNGL-PPI learns two complementary views for each protein. The global branch applies Graph Isomorphism Network (GIN) to the full PPI graph:

$$h_v^{global} = \text{GIN}(v, G, X), \quad h_v^{global} \in \mathbb{R}^{512}.$$

This branch uses message passing on the PPI network to capture the position and interaction context of protein v in the full graph.

The local branch extracts a K -hop subgraph G_v^K around each protein and applies GIN-AK:

$$h_v^{local} = \text{GIN-AK}(G_v^K), \quad h_v^{local} \in \mathbb{R}^{512}.$$

This branch captures local neighborhood structure, which is related to the subgraph-based modeling techniques discussed in the course.

The paper describes GNNGL-PPI as combining global graph and local subgraph features. In the released implementation, these two 512-dimensional node representations are fused by a fixed additive operation:

$$h_v = h_v^{global} + h_v^{local}.$$

For a protein pair (u, v) , the pair representation is computed by element-wise multiplication:

$$h_{uv} = h_u \odot h_v, \quad \hat{y}_{uv} = \sigma(\text{MLP}(h_{uv})).$$

This gives seven probabilities, one for each interaction category.

4 Proposed method: Adaptive Global–Local Fusion

The fixed additive fusion treats global and local representations as equally important for every protein and every embedding dimension. This may be restrictive because different proteins may require different levels of global and local information. For example, proteins located in clear functional clusters may rely more on global graph structure, while proteins with distinctive local neighborhoods may rely more on subgraph-level patterns.

We propose Adaptive Global–Local Fusion. The main modification is to replace fixed additive fusion with a learnable feature-wise gate:

$$\alpha_v = \sigma(\text{MLP}_{gate}([h_v^{global} \| h_v^{local}])), \quad \alpha_v \in [0, 1]^{512}.$$

The final protein representation is:

$$h_v^{final} = 2(\alpha_v \odot h_v^{global} + (1 - \alpha_v) \odot h_v^{local}).$$

The factor 2 keeps the scale comparable to the released fixed-sum baseline when $\alpha_v = 0.5$. The prediction head remains unchanged:

$$h_{uv} = h_u^{final} \odot h_v^{final}, \quad \hat{y}_{uv} = \sigma(\text{MLP}(h_{uv})).$$

For training, we will use Asymmetric Loss for label imbalance, following the implementation setting.

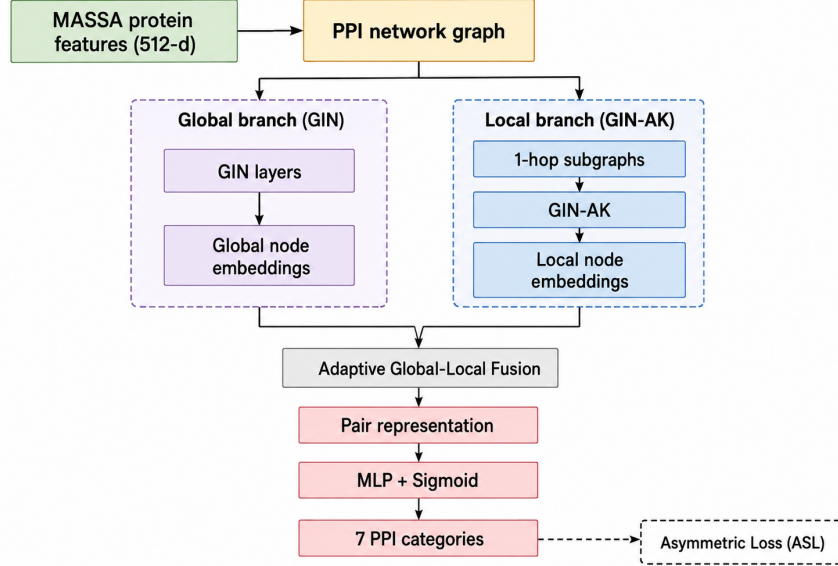


Figure 1: Overview of the proposed Adaptive GNNGL-PPI.

5 Model suitability and experimental plan

The proposed model is suitable for three reasons. First, PPI data is naturally graph-structured, so GNNs can exploit both node features and network topology. Second, the task is edge-level prediction, which matches the GNN pipeline of learning node representations and combining pairs of node embeddings for prediction. Third, GNNGL-PPI explicitly models both full-network interaction patterns and local subgraph structures, which are central sources of information in PPI networks.

We will compare the following variants and ablation settings:

Table 2: Planned model comparison and ablation studies.

Group	Variant	Purpose
Feature source	Global-only	Disable the local branch
Feature source	Local-only	Disable the global branch
Fusion strategy	Fixed-sum GNNGL-PPI	Released additive fusion
Fusion strategy	Concat-MLP fusion	Non-adaptive learnable fusion
Fusion strategy	Scalar-gate fusion	Protein-wise global-local weighting
Fusion strategy	Feature-wise adaptive fusion	Proposed model
Loss function	BCE vs. ASL	Isolate the effect of imbalance-aware loss
Subgraph size	$K = 1$ vs. $K = 2$	Test local neighborhood size if feasible

The main experiments will evaluate whether adaptive fusion improves over the released fixed-sum implementation. We will first compare global-only, local-only, and global-local representations to measure the contribution of each feature source. We will then compare fixed-sum, Concat-MLP, scalar-gate, and feature-wise adaptive fusion. Finally, we will test BCE versus ASL and optionally vary the local subgraph size if computational resources allow.

6 Expected contribution

This project aims to reproduce and analyze GNNGL-PPI for multi-category PPI prediction, then investigate whether adaptive global-local fusion can improve the fixed additive fusion used in the released implementation. The expected contribution is a lightweight and implementation-friendly extension that dynamically combines global graph features and local subgraph features for each protein, supported by ablation studies on feature source, fusion strategy, and loss function.